

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

May-June-2017

Max. Marks: 100

Class: SE

Name of the Course : Computational Mathematics

Course Code : (NON CBGS)

Duration: 3 Hrs.

Semester: IV

Branch: IT

Instructions:

(1) All Questions are Compulsory.

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Q. No		Questions										Max Marks
Q.1		Attempt any Four of the following										
	a).	The probability distribution of a random variable x is given by										24
		X	-2	-1	0	1	2	3				
		P(x)	0.1	k	0.2	2k	0.3	k				
		Find (i) k , (ii) mean, (iii) variance										
	b).	Out of 1000 families of 3 children each, how many would you expect to have 2 boys and 1 girl?										
	c).	Find mean and variance of Poisson Distribution.										
	d).	Fit a Poisson distribution to the following data										
		X	0	1	2	3	4	5	6	7	8	Total
		f	56	156	132	92	37	22	4	0	1	200
	e).	Two unbiased dice are thrown find expectation of sum.										
	f).	The probability that a man aged 60 will live up to 70 is 0.65. What is the probability that out of 10 such men now at 60 at least 7 will live up to 70?										
Q.2		Attempt any Four of the following										
	a).	A sample of 100 students is taken from a large population the mean heights of students in this sample is 160 cm. Can it be reasonably regarded that the population mean height is 165 cm. and standard deviation is 10 cm.?										28
	b).	A random sample of size 36 has 53 as mean and sum of squares of deviation from mean is 150. Can this sample be regarded as drawn from the population having 54 as mean?										
	c).	Two batches of 12 animals are taken for test of inoculation. One batch was inoculated and other was not, from the data can it be regarded as effective against the disease?										
			Dead			Survived			Total			
		Inoculated	2			10			12			
		Non-Inoculated	8			4			12			
		Total	10			14			24			
	d).	Ten individuals are chosen at random from population and their heights are found to be 63, 63, 64, 65, 66, 69, 69, 70, 71 inches. Discuss the suggestion that mean height of universe is 65 inches.										

	e).	A machine is designed to produce insulating washers for electrical devices of average thickness of 0.025 cm. A random sample of 10 washers was found to have an average thickness of 0.024 cm. with standard deviation 0.002 cm. Test the significances if deviation.								
Q.3		Attempt any Two of the following								12
	a).	Find the coefficient of correlation between the marks of a student in two tests.								
		Test-I	65	66	67	67	68	69	71	73
		Test-II	67	68	64	68	72	70	69	70
	b).	The equations of two lines of regression are $3x + 2y = 26$ and $6x + y = 31$ find (a) Means of x and y (b) Coefficients of Correlation between x and y (c) σ_y if $\sigma_x = 3$								
	c).	Fit a straight line $y = ax + b$ to the following data								
		X	1	2	3	4	5	6		
		Y	40	54	62	70	80	82		
Q4.		Attempt any Four of the following								28
	a).	Using forward difference formula find y when $x = 0.5$ from the following data.								
		X	0	1	2	3				
		Y	-1	1	1	-2				
	b).	Evaluate $\int$ by using (i) Trapezoidal rule (ii) Simpson's (1/3)rd rule. Take $h=0.25$.								
	c).	Solve: using Gauss- Jordan method $2x + y + 4z = 16$; $3x + 2y + z = 10$; $x + 3y + 3z = 16$.								
	d).	Using Newton- Raphson Method to solve following equation. $X^3 + x - 1 = 0$								
	e).	Express into factorial polynomial the function. $x^4 + 6x^3 - 6x^2 + 5x$								
Q5.		Attempt any One of the following								8
	a)	Solve the following L.P.P. by Simplex method Maximize $z = x_1 + 3x_2$ Subject to $x_1 + 2x_2 \leq 10$, $0 \leq x_1 \leq 5$, $0 \leq x_2 \leq 4$, $x_1 \geq 0$, $x_2 \geq 0$								
	b)	Find (i) all basic solution (ii) all feasible basic solution (iii) Optimal feasible basic solution for the following L.P.P. Maximize $z = 2x_1 - 3x_2 + 4x_3 - 5x_4$ Subject to $x_1 + 4x_2 - 2x_3 + 8x_4 = 2$ $-x_1 + 2x_2 + 3x_3 + 4x_4 = 1$ $x_1 \geq 0$, $x_2 \geq 0$, $x_3 \geq 0$, $x_4 \geq 0$								